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WEEK-4 PRACTICAL

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the following queries:

- 1 • `drop database if exists bookflow_db;`
- 2 • `CREATE DATABASE bookflow_db;`
- 3 • `USE bookflow_db;`
- 4 • `SELECT DATABASE();`

The Output tab shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	14:37:18	Drop database if exists bookflow_db	Error Code: 1064. You have an error in your SQL synta...	0.000 sec
2	14:37:35	drop database if exists bookflow_db	8 row(s) affected	0.203 sec
3	14:37:46	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
4	14:37:54	USE bookflow_db	0 row(s) affected	0.000 sec
5	14:38:31	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec

The SQL Editor shows the result of the last query: `DATABASE()` returning `bookflow_db`.

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

MySQL Model x

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

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- Client Connections
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- Data Import/Restore

INSTANCE

- Startup / Shutdown
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Administration Schemas

Information

No object selected

Week_1_practical

SQL File 3*

SQL File 4*

Limit to 1000 rows

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

13 genre VARCHAR(100),

14

15 description TEXT,

16

17 published_year INT

18);

19 describe Books;

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

Result Grid

Form Editor

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	

Result 2 x

Read Only

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	14:37:35	drop database if exists bookflow_db	8 row(s) affected	0.203 sec
3	14:37:46	CREATE DATABASE bookflow_db	1 row(s) affected	0.000 sec
4	14:37:54	USE bookflow_db	0 row(s) affected	0.000 sec
5	14:38:31	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
6	14:39:13	CREATE TABLE Books (book_id INT AUTO_INC...	0 row(s) affected	0.047 sec
7	14:39:55	describe Books	6 row(s) returned	0.015 sec / 0.000 sec

Object Info

Session

Query Completed

MySQL Workbench

MySQL Model x

Local instance MySQL80 x

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

Week_1_practicalSQL File 3*SQL File 4* x

SQLAdditions

MANAGEMENT

Server Status

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Server Logs

AdministrationSchemas

Information

No object selected

112author,

113genre,

114published_year

115FROM Books

116ORDER BY book_id;

117

118

Result Grid

Filter Rows:

Edits

Export/Import:

Result Grid

Form Editor

book_idtitleauthorgenre

1The MartianAndy WeirScience Fiction

2DuneFrank HerbertScience Fiction

3The Hitchhikers Guide to the GalaxyDouglas AdamsScience Fiction

4Pride and PrejudiceJane AustenRomance

5The Great GatsbyF. Scott FitzgeraldClassic

6FoundationIsaac AsimovScience Fiction

7The HobbitJ.R.R. TolkienFantasy

Books 3 x

Apply

Context Help

Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	14:37:54	USE bookflow_db	0 row(s) affected	0.000 sec
5	14:38:31	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec
6	14:39:13	CREATE TABLE Books (book_id INT AUTO_INC...	0 row(s) affected	0.047 sec
7	14:39:55	describe Books	6 row(s) returned	0.015 sec / 0.000 sec
8	14:40:52	INSERT INTO Books (title, author, genre, ...	10 row(s) affected Records: 10 Duplicates: 0 Wami...	0.032 sec
9	14:41:19	SELECT book_id, title, author, genre, pu...	10 row(s) returned	0.000 sec / 0.000 sec

Object Info

Session

Query Completed

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MySQL Workbench

MySQL Model xLocal instance MySQL80 x

FileEditViewQueryDatabaseServerToolsScriptingHelp

Navigator

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Information

Week_1_practicalSQL File 3*SQL File 4*

Limit to 1000 rows

SQLAdditions

113genre,
114published_year
115FROM Books
116ORDER BY book_id;
117ALTER TABLE Books
118ADD COLUMN embedding JSON;
119DESCRIBE Books;
120

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

No object selected

Result Grid

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	
embedding	json	YES		NULL	

Result GridForm Editor

Result 4 xRead OnlyContext HelpSnippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	14:41:19	SELECT book_id, title, author, genre, publi...	10 row(s) returned	0.000 sec / 0.000 sec
10	14:43:29	ALTER TABLE Books ADD COLUMN embedding JSON	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.031 sec
11	14:44:04	DESCRIBE Books	7 row(s) returned	0.000 sec / 0.000 sec

Object InfoSession

Query Completed

MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator Week_1_practical SQL File 3* SQL File 4* SQLAdditions

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INSTANCE

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Administration

Information

No object selected

Result Grid Filter Rows: Edit: Export/Import: Result Grid Form Editor

```
149 WHERE book_id = 10;
150 SELECT
151     book_id,
152     title,
153     embedding
154 FROM Books
155 ORDER BY book_id;
156
```

book_id	title	embedding
1	The Martian	[0.90, 0.80, 0.20]
2	Dune	[0.85, 0.75, 0.25]
3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
4	Pride and Prejudice	[0.10, 0.20, 0.90]
5	The Great Gatsby	[0.15, 0.30, 0.80]
6	Foundation	[0.95, 0.65, 0.20]
7	The Hobbit	[0.30, 0.75, 0.90]
8	Enders Game	[0.88, 0.85, 0.15]

Books 5 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 20	14:50:57	UPDATE Books SET embedding = JSON_ARRAY(0.20...	1 row(s) affected Rows matched: 1 Changed: 1 Wami...	0.015 sec
✓ 21	14:51:08	UPDATE Books SET embedding = JSON_ARRAY(0.70...	1 row(s) affected Rows matched: 1 Changed: 1 Wami...	0.016 sec
✓ 22	14:51:19	SELECT book_id, title, embedding FROM Book...	10 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Week_1_practical SQL File 3* SQL File 4* SQLAdditions

MANAGEMENT

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Administration

No object selected

Information

Result Grid

	book_id	title	author	genre	l2_distance
1	1	The Martian	Andy Weir	Science Fiction	0
8	8	Enders Game	Orson Scott Card	Science Fiction	0.0735
2	2	Dune	Frank Herbert	Science Fiction	0.0866

Result 6 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
21	14:51:08	UPDATE Books SET embedding = JSON_ARRAY(0.70...	1 row(s) affected Rows matched: 1 Changed: 1 Wami...	0.016 sec
22	14:51:19	SELECT book_id, title, embedding FROM Book...	10 row(s) returned	0.000 sec / 0.000 sec
23	14:56:26	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.016 sec / 0.000 sec

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Week_1_practical SQL File 3* SQL File 4* x SQLAdditions

MANAGEMENT

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Administration

Information

No object selected

Limit to 1000 rows

FROM Books

WHERE embedding IS NOT NULL

ORDER BY cosine_similarity DESC

LIMIT 3;

Result Grid

	book_id	title	author	genre	cosine_similarity
1	1	The Martian	Andy Weir	Science Fiction	1
2	2	Dune	Frank Herbert	Science Fiction	0.9986
8	8	Enders Game	Orson Scott Card	Science Fiction	0.9983

Result 7 x Read Only Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
22	14:51:19	SELECT book_id, title, embedding FROM Book...	10 row(s) returned	0.000 sec / 0.000 sec
23	14:56:26	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.016 sec / 0.000 sec
24	15:03:59	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.000 sec / 0.000 sec

Query Completed

Explorer

BOOKFLOW_API

- node_modules
- package-lock.json
- package.json

package.json

```
{
  "name": "bookflow-api",
  "version": "1.0.0",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "dependencies": {
    "express": "5.1.0",
    "mysql2": "3.14.3"
  }
}
```

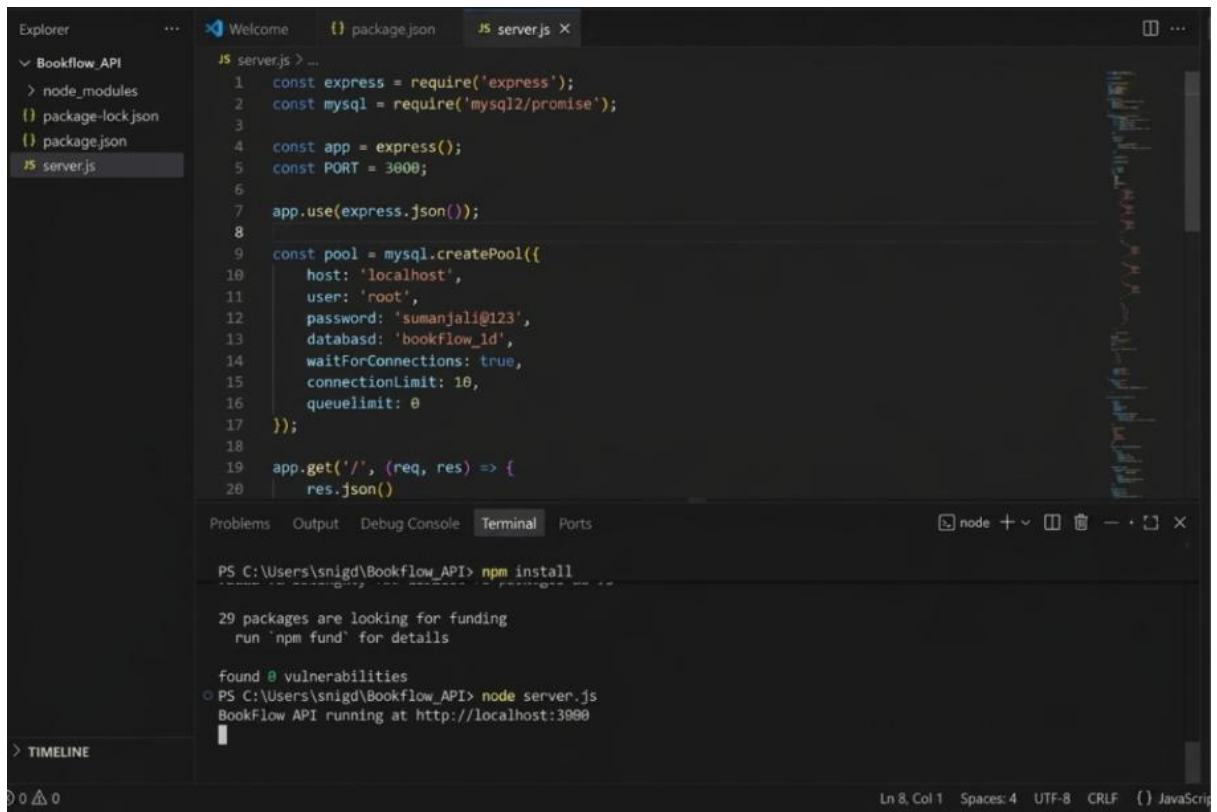
Terminal

PS C:\Users\snigd\Bookflow_API> npm install

added 78 packages, and audited 79 packages in 9s

29 packages are looking for funding
run 'npm fund' for details

found 0 vulnerabilities



```
server.js > ...
1  const express = require('express');
2  const mysql = require('mysql2/promise');
3
4  const app = express();
5  const PORT = 3000;
6
7  app.use(express.json());
8
9  const pool = mysql.createPool({
10     host: 'localhost',
11     user: 'root',
12     password: 'sumanjali@123',
13     databasd: 'bookflow_id',
14     waitForConnections: true,
15     connectionLimit: 10,
16     queueLimit: 0
17 });
18
19 app.get('/', (req, res) => {
20     res.json()
```

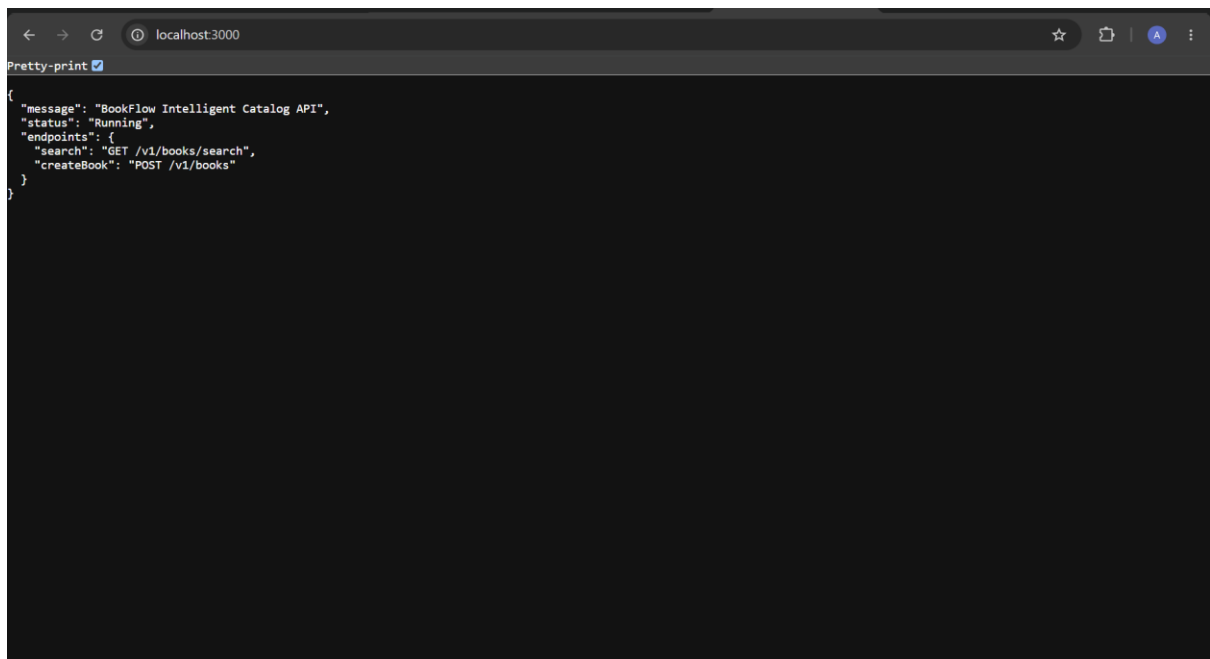
PS C:\Users\sngid\Bookflow_API> npm install

29 packages are looking for funding
run 'npm fund' for details

found 0 vulnerabilities

PS C:\Users\sngid\Bookflow_API> node server.js

BookFlow API running at http://localhost:3990



Swagger Editor
Powered by SMARTIDEAS

File

Edit

Generate Server

Generate Client

About

Sign in to Swagger

Try Swagger

1openapi: 3.0.3

2

3info:

4 title: BookFlow Intelligent Catalog API

5 version: 1.0.0

6 description: |

7 REST API for semantic book search.

8

9 The API searches books using mock

10 three-dimensional vectors stored in MySQL.

11

12servers:

13 - url: http://localhost:3000

14

15tags:

16 - name: Books

17 description: Book operations

18

19paths:

20

21 /v1/books/search:

22 get:

23 tags:

24 - Books

25

26 summary:

27 Search books using semantic similarity

28

29 parameters:

30

31 - name: q

BookFlow Intelligent Catalog API

1.0.0OAS 3.0

REST API for semantic book search.

The API searches books using mock three-dimensional vectors stored in MySQL.

Servers

http://localhost:3000

Books

Book operations

GET

/v1/books/searchSearch books using semantic similarity

POST

/v1/booksCreate a new book

Schemas

Curl

curl -X 'GET' \
 'http://localhost:3000/v1/books/search?q=Stories%20about%20space&limit=3' \
 -H 'accept: application/json'

Request URL

http://localhost:3000/v1/books/search?q=Stories%20about%20space&limit=3

Server response

Code	Details
200	Response body <pre>{ "description": "An astronaut struggles to survive alone on Mars.", "published_year": 2011, "similarity": 1 }, { "book_id": 2, "title": "Dune", "author": "Frank Herbert", "genre": "Science Fiction", "description": "A story involving planets, politics, survival and space.", "published_year": 1965, "similarity": 0.9986 }, { "book_id": 8, "title": "Enders Game", "author": "Orson Scott Card", "genre": "Science Fiction", "description": "A young student is trained for an interstellar conflict." }</pre>

Swagger Editor

File Edit Generate Server Generate Client About

Sign in to Swagger Try Swagger

```
1 openapi: 3.0.3
2
3 info:
4   title: BookFlow Intelligent Catalog API
5   version: 1.0.0
6   description: |
7     REST API for semantic book search.
8
9   The API searches books using mock
10  three-dimensional vectors stored in MySQL.
11
12 servers:
13   - url: http://localhost:3000
14
15 tags:
16   - name: Books
17     description: Book operations
18
19 paths:
20   /v1/books/search:
21     get:
22       tags:
23         - Books
24       summary:
25         Search books using semantic similarity
26       parameters:
```

Request URL

http://localhost:3000/v1/books

Server response

Code Details

201

Response body

```
{
  "message": "Book created successfully",
  "book": {
    "book_id": 11,
    "title": "Project Hail Mary",
    "author": "Andy Weir",
    "genre": "Science Fiction",
    "description": "A science fiction survival story involving spac...",
    "published_year": 2021,
    "embedding": [
      0.03,
      0.02,
      0.12
    ]
  }
}
```

Response headers

```
content-length: 253
content-type: application/json; charset=utf-8
```

Responses

Code	Description	Links
201	Book created successfully	No links

MySQL Workbench

MySQL Model Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

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Administration

Information

No object selected

Week_1_practical SQL File 3* SQL File 4*

Limit to 1000 rows

SQL

```
311 SELECT
312     book_id,
313     title,
314     author,
315     genre,
316     published_year
317 FROM Books
318 WHERE title = 'Project Hail Mary';
319
```

Result Grid

book_id	title	author	genre	published_year
11	Project Hail Mary	Andy Weir	Science Fiction	2021
NULL	NULL	NULL	NULL	NULL

Books 8 x Apply Revert Context Help Snippets

Output

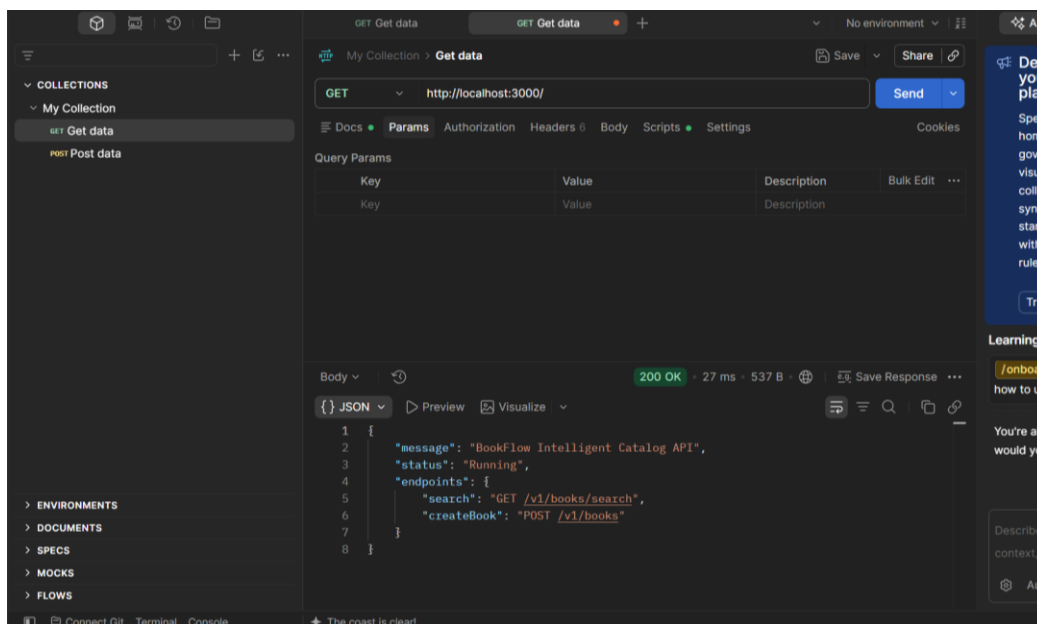
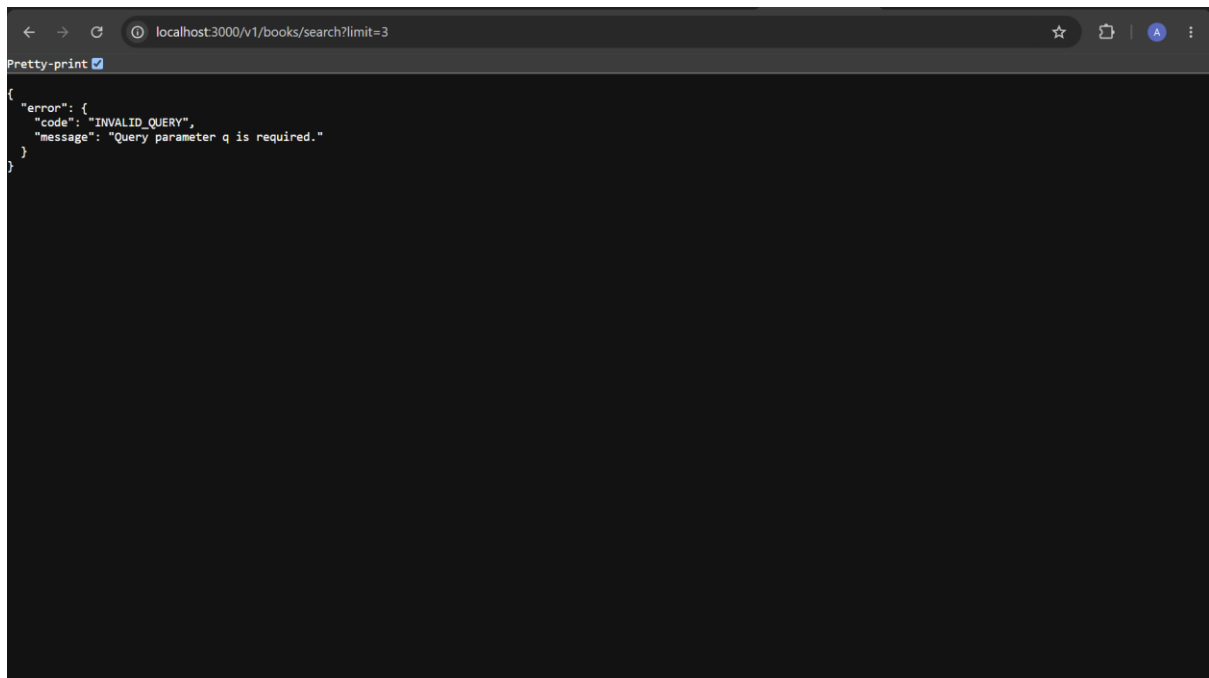
Action Output

#	Time	Action	Message	Duration / Fetch
23	14:56:26	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.016 sec / 0.000 sec
24	15:03:59	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.000 sec / 0.000 sec
25	21:30:47	SELECT book_id, title, author, genre, publi...	1 row(s) returned	0.110 sec / 0.000 sec

Object Info Session

Query Completed

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GET Get data

My Collection > Get data

GET http://localhost:3000/v1/books/search?q=Stories%20about%20space&limit=3

Query Params

Key	Value	Description	Bulk Edit	...
q	Stories%20about%20space			
limit	3			

Body

200 OK - 78 ms - 1.01 KB

```
{
  "query": "Stories about space",
  "count": 3,
  "results": [
    {
      "book_id": 1,
      "title": "The Martian",
      "author": "Andy Weir",
      "genre": "Science Fiction",
      "description": "An astronaut struggles to survive alone on Mars.",
      "published_year": 2011,
      "similarity": 1
    }
  ]
}
```

GET Get data

My Collection > Get data

GET http://localhost:3000/v1/books/search?limit=3

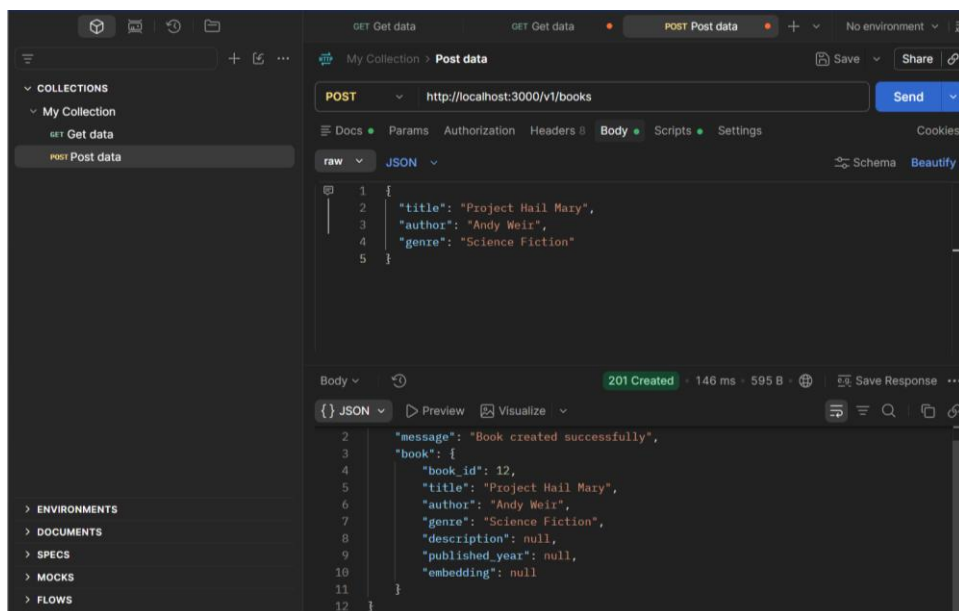
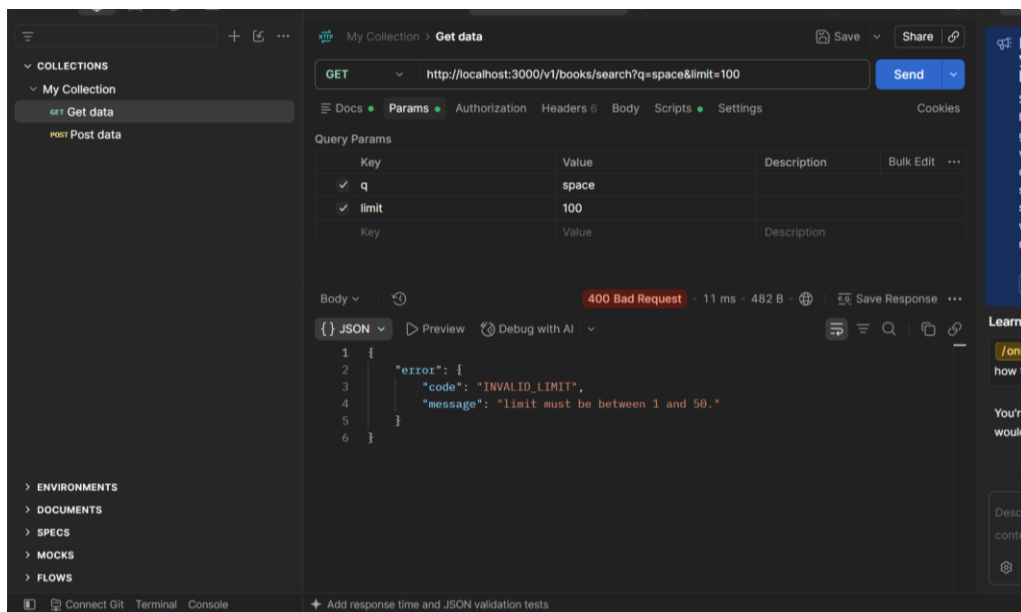
Query Params

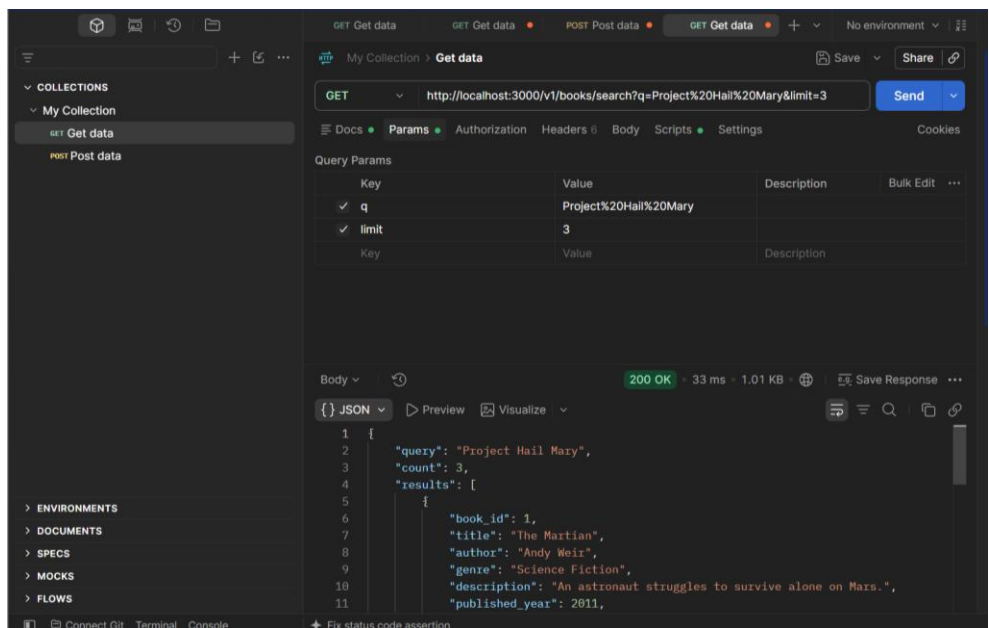
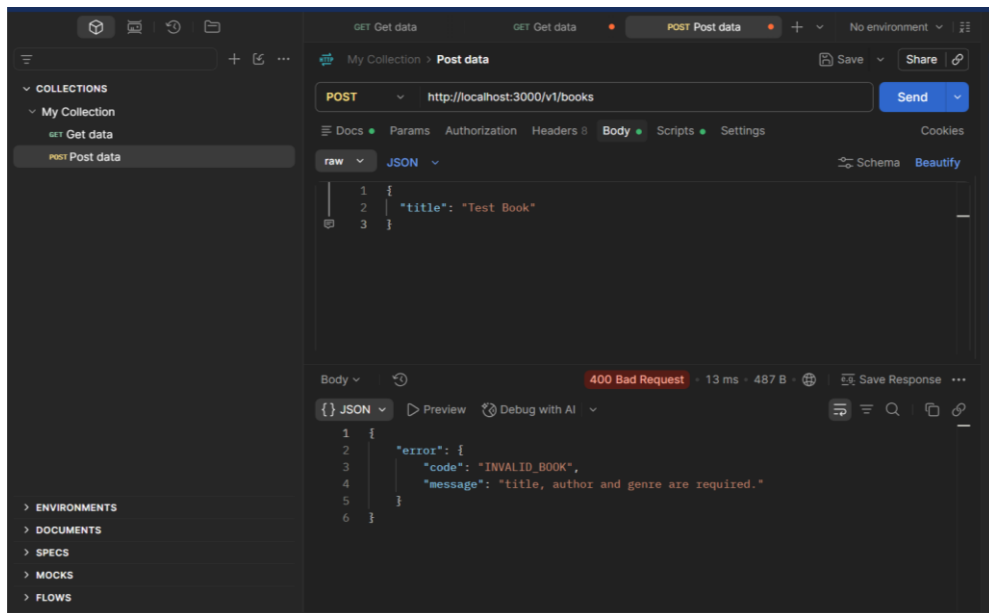
Key	Value	Description	Bulk Edit	...
limit	3			

Body

400 Bad Request - 7 ms - 481 B

```
{
  "error": {
    "code": "INVALID_QUERY",
    "message": "Query parameter q is required."
  }
}
```





MySQL Workbench

MySQL Model x Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator Week_1_practical SQL File 3* SQL File 4* x SQLAdditions

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Administration

Information

No object selected

Limit to 1000 rows

```
320 book_id,  
321 title,  
322 author,  
323 genre,  
324 published_year,  
325 embedding  
326 FROM Books  
327 ORDER BY book_id;  
328
```

Result Grid

book_id	title	author	genre	published_year
7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
8	Enders Game	Orson Scott Card	Science Fiction	1977
9	Harry Potter and the Sorcerer's Stone	J.K. Rowling	Fantasy	1997
10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870
11	Project Hail Mary	Andy Weir	Science Fiction	2021
12	Project Hail Mary	Andy Weir	Science Fiction	2021
NULL	NULL	NULL	NULL	NULL

Books 9 x Apply Revert Context Help Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 24	15:03:59	SELECT book_id, title, author, genre, R...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 25	21:30:47	SELECT book_id, title, author, genre, publi...	1 row(s) returned	0.110 sec / 0.000 sec
✓ 26	22:53:36	SELECT book_id, title, author, genre, publi...	12 row(s) returned	0.015 sec / 0.000 sec

Object Info Session

Query Completed

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.